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Why Keys are Needed

Keys solve many fundamental database problems.

Problem 1 — Duplicate Records

Without keys	With a key
The database cannot differentiate.	Now every student is uniquely identifiable.

Problem 2 — Fast Searching

Without a key	With a key
finding one student requires checking every row.	the DBMS can use indexes for extremely fast lookups.

Problem 3 — Updating Records

Suppose you want to update Rahul's phone number.

Without a key	With a key
`UPDATE Student SET Phone='9999999999' WHERE Name='Aadya';`	`UPDATE Student SET Phone='9999999999' WHERE RollNo=006;`
Oops. **Maybe 50 aadya get updated.**	**Only one row changes.**

Problem 4 — Deleting Records

Without keys	With keys
`DELETE FROM Student WHERE Name='Aadya';`	`DELETE FROM Student WHERE RollNo=006;`
Danger. **Multiple students may disappear.**	**Exactly one row gets deleted.**

Without keys, relationships become impossible.

Uniqueness

No two rows can have the same key value.
Every table must have a primary key, and its value can never be NULL.

Why?
Because every row must be identifiable.

> **Primary Key = Unique + NOT NULL** , both these conditions are required for entity integrity , Without Entity Integrity

- duplicate identities
- missing identities
- impossible updates
- impossible deletions

6. Referential Integrity

This integrity rule deals with relationships between tables.
Suppose

Student

RollNo	Name
101	arpit
102	aadya

Fees

Receipt	RollNo
1	101
2	105

Problem
Student 105 doesn't exist.
The Fees table now references a nonexistent student.
This is called an orphan record.

Referential Integrity says
Every foreign key value must either

- match an existing primary key
  OR
- be NULL (if allowed) ,

Normalization

Normalization removes redundancy and improves database design.
Keys are essential for normalization.
Without keys,
the DBMS cannot determine

- uniqueness
- dependencies
- relationships
- decomposition rules.

keys are used for:

- Identification
- Relationships
- Constraints
- Indexing
- Normalization
- Data integrity , refernecial integrity
- Updates
- Deletes